Waterbench

Towards Democratizing Secure Image Authorship with Watermarking

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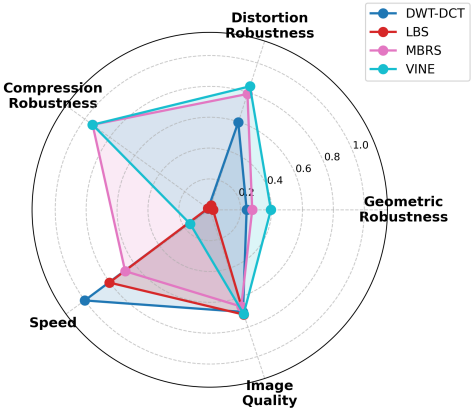
Jan Schmidt & Fabio Genz



Outline

1 Introduction

2 Background

3 Benchmarking Setup

4 Results

5 Demo & Conclusion

Motivation & Research Question

The Problem:

- AI-generated images are indistinguishable from real photographs
- Content is shared without attribution
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How can we democratize cryptographically-verified image ownership through accessible watermarking technology?

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Our Contributions

1. First **end-user-oriented** benchmark of invisible watermarking methods
2. **Waterbench** platform for accessible image ownership

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Watermarking Taxonomy

In-Processing (out of scope)

- Embed during image *generation* (e.g., Stable Signature [3], Tree-Ring [4])
- Limited to AI-generated content

Post-Processing (our focus)

- Applied to *any* existing digital image
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Domain	Method	Key Idea
Spatial	LSB [5]	Modify least significant bit
Transform	DWT-DCT-SVD [6]	Embed in frequency coefficients
Deep Learning	MBRS [7]	Mini-batch JPEG training
	RAW [8]	Zero-bit detection
	VINE [2]	Diffusion model priors

Table: Five benchmarked post-processing methods.

Classical Methods: LSB & DWT-DCT-SVD

LSB – Least Significant Bit [5]

- Directly modifies the least significant bits of pixel values
- **+** Very fast, high embedding capacity
- **–** Extremely fragile: fails under basic JPEG compression or filtering
- Serves as an important *baseline*

Example: Pixel value 142 (10001110_2) $\rightarrow$ embed bit '1' $\rightarrow$ 143 (10001111_2)

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DWT-DCT-SVD [6]

- Embeds watermark in frequency domain using:
  - **DWT**: spatial-frequency decomposition into sub-bands
  - **DCT**: energy compaction in mid-frequency coefficients
  - **SVD**: singular values are stable under perturbations
- **+** Moderate robustness against filtering and JPEG
- **–** Struggles with geometric attacks (rotation, crop)

Deep Learning Methods

Method	Key Innovation
MBRS [7]	Noiser randomly switches between *real* JPEG, *simulated* JPEG, and noise-free layers per mini-batch
VINE [2]	Uses pretrained diffusion model priors; blurring as surrogate for image editing attacks
RAW [8]	Zero-bit detection via learnable watermark + classifier (no encoder–decoder) $\Rightarrow$ excluded from primary comparison

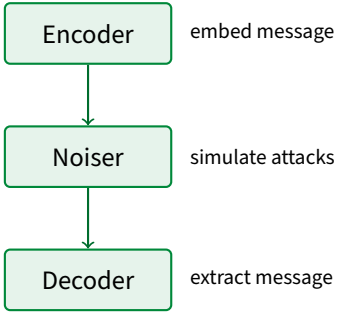

Figure: Encoder–noiser–decoder architecture (MBRS, VINE). Trained end-to-end to reconstruct the embedded message.

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Evaluation Setup

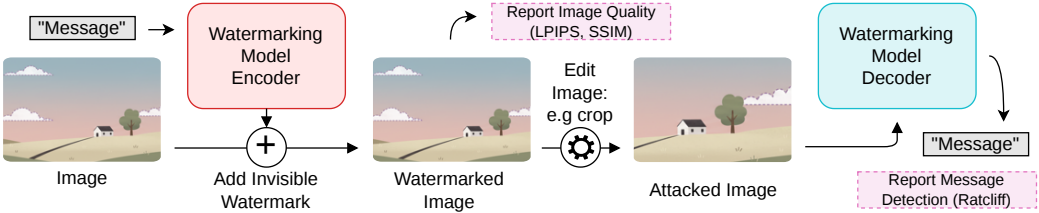

Figure: Our benchmarking pipeline follows a three step process.

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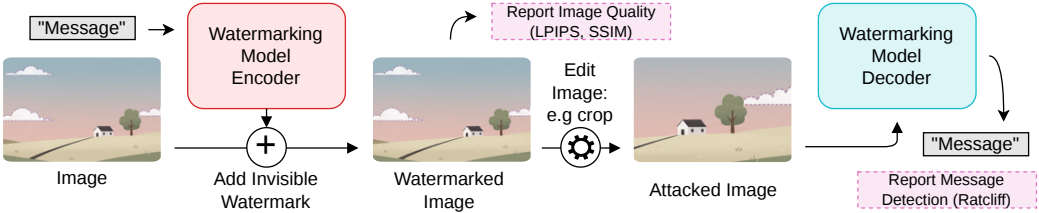

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Configuration:

- **100 images** per configuration (watermarking method, attack type and message length)
- **8 attack types**, with varying magnitudes (rotation, scaling, cropping, brightness, contrast, blur, noise, JPEG compression)
- **5 message lengths**: 2, 4, 8, 15, 30 ASCII characters

Metrics & Evaluation Dimensions

Message Reconstruction – Ratcliff/Obershelp [9]:

$$RO = \frac{2|K_m|}{|S_1| + |S_2|}$$

- S_1, S_2: original and extracted message
- $|K_m|$: sum of lengths of all common substrings
- Position-sensitive (unlike Jaccard [10])
- Score: 0 (no match) to 1 (identical)

Five Evaluation Dimensions:

Dimension	Definition
Distortion Rob.	Mean Ratcliff across brightness, contrast, blur, noise
Geometric Rob.	Mean Ratcliff across rotation, scaling, cropping
Compression Rob.	Mean Ratcliff across JPEG quality levels
Image Quality	Mean SSIM & LPIPS (original vs. watermarked)
Speed	Normalized processing efficiency (fastest = 1.0)

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Robustness & Image Quality Results

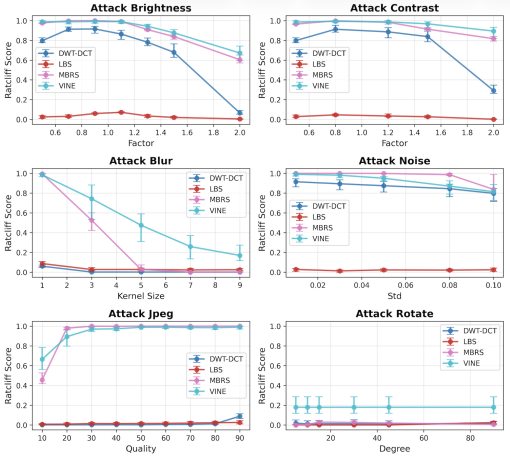

Figure: Robustness across attack types and magnitudes. Mean Ratcliff scores averaged across message lengths.

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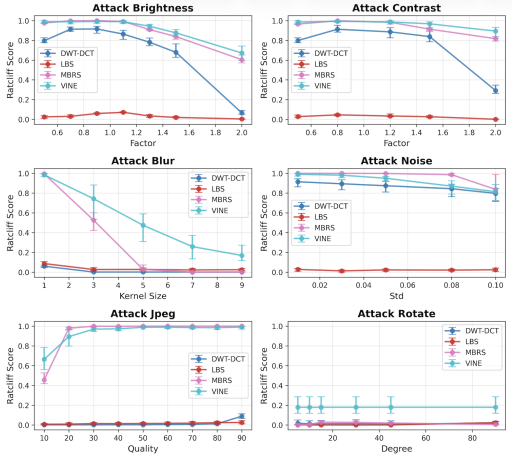

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Image Quality (no attack):

Method	LPIPS ↓	SSIM ↑
DWT-DCT	0.10 ± 0.07	0.96 ± 0.01
LSB	0.00 ± 0.00	1.00 ± 0.00
MBRS	0.23 ± 0.06	0.88 ± 0.05
RAW	0.03 ± 0.02	0.94 ± 0.02
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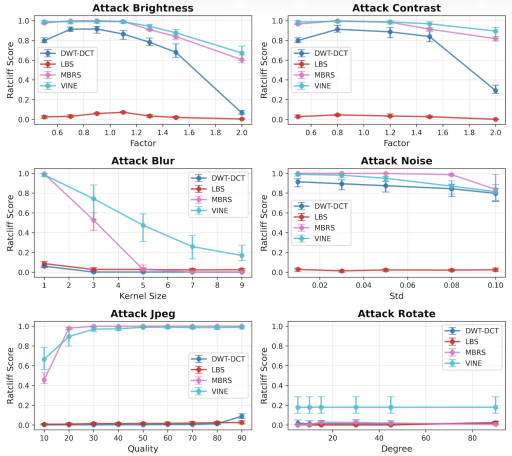

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Key Findings:

- VINE & MBRS dominate across distortion and compression attacks
- LSB fails even under minimal modifications
- **All methods struggle with geometric attacks** (scores < 0.40)

Practical Trade-off: Robustness vs. Speed

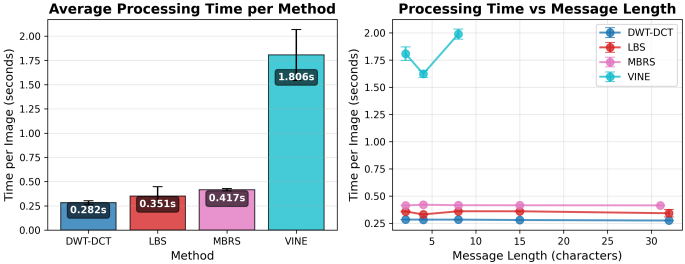

Figure: Processing time per image across methods.

	Distortion	Compression	Time/img
MBRS	0.79	**0.94**	**0.42s**
VINE	**0.84**	**0.94**	1.81s

All methods have a comparable pipeline latency except VINE. ⇒ **MBRS** = best practical trade-off for end users

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Waterbench – `waterbench.org`

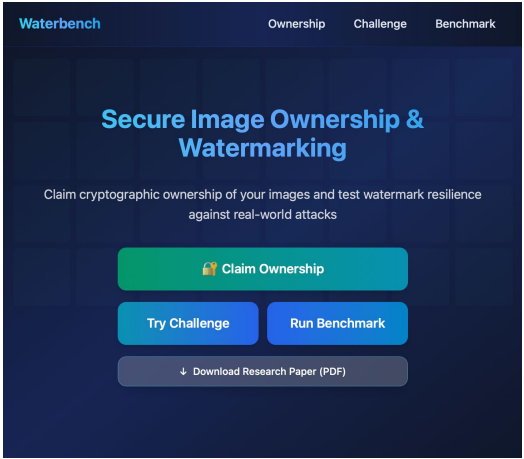


Figure: Waterbench landing page.

Ownership Mode:

- RSA-2048 keypair generation (client-side)
- Cryptographic authorship verification
- Invisible embedding & public verification

Challenge Mode:

- Interactive robustness testing
- Feedback on watermark survival

$\Rightarrow$ **Live Demo**

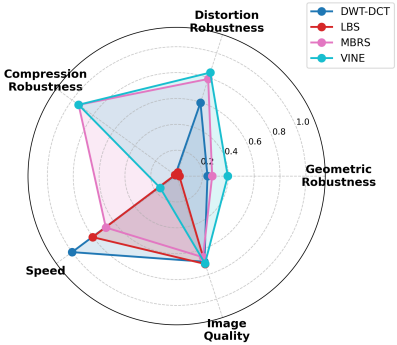
Summary & Outlook

Key Findings:

- **MBRS** offers the best trade-off: high compression robustness (0.94) at 0.42s/image
- **VINE** achieves highest overall robustness but is 4$\times$ slower
- **Geometric robustness remains unsolved** across all evaluated methods (scores < 0.40)

Limitations & Future Work:

- Benchmark limited to 100 images, 5 message lengths, and only isolated attacks
- Future: compound and real-world attack scenarios (e.g., screenshotting, social media re-encoding)
- Future: larger-scale evaluation and integration of geometric-invariant techniques into Waterbench



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